Zhaoxiang Feng

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Seeking Software Engineering (ML Systems, LLM Inference) and ML Research Internships | Available May–Aug. 2027

Education

The University of Texas at Austin — Austin, TX
Master of Science in Computer Science — *Aug. 2026 – Present*
Coursework: Post-Training of LLMs, ML for Systems, Natural Language Processing, Mathematics in Deep Learning

University of California San Diego — San Diego, CA
B.S. in Data Science; B.S. in Probability & Statistics; GPA: 3.84/4.00 — *Sep. 2022 – Mar. 2026*

Research Experience

Hao AI Lab, UC San Diego — San Diego, CA
Research Assistant, Advisor: Prof. Hao Zhang — *Sep. 2025 – Present*

- Led **JetSpec++**, a parallel-drafting speculative decoding method: **1,361 tokens/s** single-request on GB200, up to **7.45**× over autoregressive decoding and **1.54**× over DSpark
- Built JetSpec's standalone **LLM inference engine** (**paged KV cache**, continuous batching, **Triton** tree attention, **CUDA-graph** verification): up to **1,150 tokens/s** single-stream on one B200
- Co-developed **JetSpec**, which drafts a full token tree in one forward pass: up to **9.64**× end-to-end speedup over autoregressive decoding (Qwen3-8B, MATH-500, H100)

Q-Lab, UC San Diego — San Diego, CA
Research Assistant, Advisor: Prof. Lianhui Qin — *Jan. 2025 – Jun. 2026*

- Contributed to **ArcMemo** (**Runner-Up, ARC Prize 2025 Paper Awards**): its composable memory of reusable reasoning concepts and its reasoning-guided concept selection
- Built a data pipeline that turns hand-written ARC concepts into executable puzzle generators and validated puzzles, adapting the open-source BARC generator

Projects

DFlash on Google Cloud TPU | *JAX, vLLM tpu-inference* — Code | Google Developers Blog

- Ported DFlash speculative decoding from GPU to TPU in JAX with the Google Cloud TPU team, from a standalone draft-verify loop to vLLM serving integration
- Upstreamed a merged **2.8K-line PR** into vLLM tpu-inference (#1868): the DFlash draft model, its attention layer, JAX and torchax proposers, and unit tests
- Reached **3.13**× **average speedup** over nine benchmarks on TPU v5p (Qwen3-4B), up to 5.72× on MATH-500
- Reached **2.29**× end-to-end vLLM serving speedup with Llama-3.1-8B, vs. 1.30× for EAGLE-3
- Showed that TPU verification cost stays flat as draft block size grows to **1,024**, so wider blocks are nearly free

LLM Pre-training System | *PyTorch, torch.compile, DDP, ZeRO-1, FlashAttention-2* — Sep. 2025 – Dec. 2025

- Built the training stack for a **1.8B-parameter** model on **8 NVIDIA B200 GPUs** and pre-trained it on **117B tokens in 55 hours** (about **630K tokens/s**, **36% model FLOPs utilization**)
- Implemented supervised fine-tuning (**SFT**) and **DPO** preference tuning on the Alpaca and HH-RLHF datasets

Publications

Z. Feng, L. Hu (equal contribution), et al. JetSpec++: Foresight Parallel Drafting with Self Refinements. arXiv:2610.19792, 2026. Under review at ICLR 2027.

L. Hu, **Z. Feng**, et al. JetSpec: Breaking the Scaling Ceiling of Speculative Decoding with Parallel Tree Drafting. arXiv:2606.18394, 2026. Under review at ICLR 2027.

M. Ho, C. Si, **Z. Feng**, et al. ArcMemo: Abstract Reasoning Composition with Lifelong LLM Memory. arXiv:2509.04439, 2025.

Technical Skills

Languages: Python, SQL, Java, R, JavaScript
ML Systems: PyTorch, JAX, Triton, vLLM, Hugging Face Transformers, DDP, ZeRO-1, CUDA graphs
Hardware & Tools: NVIDIA B200/GB200/H100, Google Cloud TPU v4/v5p/v6e, Git, Linux, Docker